

Deepak Roshan A

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PROFESSIONAL SUMMARY

AI Engineer specializing in **Python**, **Machine Learning**, and **AWS**, with hands-on experience building and deploying **end-to-end AI systems**. Developed production-like applications involving **computer vision**, **NLP**, and real-time data pipelines. Skilled in model development, backend integration, and cloud deployment, delivering scalable solutions for real-world problems.

EDUCATION

Sri Krishna College of Technology **B.E. in Artificial Intelligence and Machine Learning** | 2022 – 2026 CGPA: 7.0

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, SciPy)
AI/ML: TensorFlow, OpenCV, CNNs, Transformers
Backend: FastAPI, Flask, REST APIs, Socket.IO
Cloud: AWS (EC2, Lambda, CloudWatch, SNS), Docker, Hugging Face Spaces
Databases: PostgreSQL (Supabase), Cloudinary
Systems & Tools: Linux, System Monitoring, GitHub Actions (CI/CD)

CERTIFICATIONS

AWS Educate – Cloud Computing Foundations | Amazon Web Services 2025
Claude Code in Action | Anthropic Academy 2026
Introduction to Computer Vision with OpenCV | Coursera / OpenCV.org 2024

WORK EXPERIENCE

Cloud Operations Intern | Appin Technology Apr 2025 – Jun 2025

- Deployed applications on AWS (EC2, S3, Lambda) using Linux.
- Used shell scripts to automate basic deployment and system tasks.
- Monitored systems and logs using cron, htop.
- Optimized cloud resources for better performance.

TECHNICAL PROJECTS

Findora – AI-Powered Lost and Found Platform | React, FastAPI, MobileNetV3, MiniLM – GitHub | Live Demo 2024

- Built a full-stack lost and found platform using image, text, and location data.
- Used MobileNetV3 and MiniLM for item matching across vision and text features.
- Designed a scoring system to rank matches and trigger alerts for high-confidence results.
- Deployed using React, FastAPI, Supabase, and email notifications.

AI-Powered ISL Translation with Emotion Awareness | TensorFlow, MediaPipe, Flask – GitHub | Live Demo 2025

- Built a real-time ISL translation system using MediaPipe and a TensorFlow CNN with ~95% accuracy.
- Implemented emotion detection using facial landmarks with temporal smoothing.
- Designed a cloud-compatible system for real-time processing and deployment.
- Developed a speech-to-sign avatar using Three.js and speech recognition.

CloudGuard AI – Intelligent Cloud Monitoring & Self-Healing | AWS, Python, ML, Streamlit – GitHub | Live Demo 2025

- Monitored real-time EC2 metrics using CloudWatch.
- Used Isolation Forest to detect anomalies in system behavior.
- Automated self-healing using AWS Lambda and SNS alerts.
- Built a Streamlit dashboard for metrics, anomalies, and recovery.

VOXAI – Neural Voice Clone Studio | Python, FastAPI, ChatterboxTTS – GitHub | Live Demo 2025

- Built a voice cloning system that generates speech from short audio samples.
- Used ChatterboxTTS for zero-shot voice synthesis with accent preservation.
- Designed an audio processing pipeline for quality enhancement.
- Deployed using FastAPI with a web interface on Hugging Face Spaces.